

Energy density is the #1 leverage in the model.

12

CONNECTIONS

One node touches more systems than any other in the Global Systems Tensor Network. Verified against the model's own edge list.

energy density → other system nodes

vs next node: **climate (11)** most of any node

MOST-CONNECTED NODE

12 / 14

touches every node except AI · minerals

Energy density has 12 direct edges in the model's 62-edge graph¹ — more than any other node. It propagates to climate, food, finance, and supply chains at once, echoing IPCC AR6's framing of energy as the pivot of the transition².

repo index.html · IPCC AR6 WG3 Ch.6

HIGH COUPLING STRENGTH

 $\Sigma w = 46$

sum of edge weights (scale 1–5)

Average coupling is 3.83 / 5 — the highest hub total in the graph¹. Two edges hit max weight ($w=5$): oil & gas (incumbent transport dominance⁵) and ammonia (the H_2 carrier fix Smil argues the density ladder still demands⁶).

repo index.html · Smil 2022 · IEA WEO 2024

WHY IT'S A LEVER

Stocks + Flows

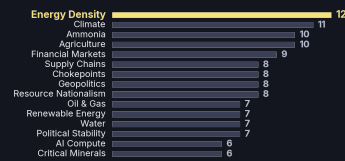
Meadows leverage rank #3 of 12

Energy density sits in Meadows' 'structure of stocks and flows'³ — the 3rd-deepest of 12 leverage classes, recently re-mapped by Abson et al. (2017)⁴. Changing carrier chemistry rewires physical infrastructure, not just prices.

Meadows 1999 · Abson et al. 2017

NODES RANKED BY CONNECTIONS (15 total)

repo index.html · EDGES array



ENERGY DENSITY OUTGOING EDGES (12 targets, $w=1-5$)

repo index.html · edge weights · type



WHY ENERGY DENSITY IS THE HIGHEST-LEVERAGE NODE

Leverage = connections × coupling strength. Energy density has **12 edges** (the most) and $\Sigma w = 46$ of coupling weight (the highest hub total)¹. Fossil fuels still deliver **59.1%** of global electricity⁷, and road-transport electrification is only projected to climb from 4% to 17% of energy use by 2030⁵. So changing energy density doesn't change one outcome — it reshapes climate, agriculture, finance, supply chains, and geopolitics in the same move. That is Meadows' third-deepest class: rewiring material stocks and flows³. **One node, many downstream fixes.**

SOURCES

¹ Global Systems Tensor Network — repo (15 nodes, 62 edges). [github.com / StewAlexander-com](https://github.com/StewAlexander-com)

² IPCC AR6 WG3, Ch.6 — Energy Systems (2022). ipcc.ch/ar6/wg3/chapter-6

³ Meadows, D. — Leverage Points (1999). donellameadows.org

⁴ Systems Thinking Alliance (2025) — Abson et al. taxonomy.systemsthinkingalliance.org

⁵ IEA, World Energy Outlook 2024 — road EVs 4→17% by 2030. [iea.org / WEO 2024](https://iea.org/WEO2024)

⁶ Smil, V. — How the World Really Works (2022); kerosene ≈ 40× Li-ion energy density.

principus.si

⁷ Ember, Global Electricity Review 2025 — fossils = 59.1% of electricity. ember-energy.org